

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA19-SS-0000000046
Award Program	2019 Seeding Solutions: Addressing Challenges in Food and Agriculture Funding Opportunity
Project Title	Genome Editing Wheat for Durable and Sustainable Disease Resistance
Grant Start Date	01/01/2020
Grant End Date	12/31/2023
Total Grant Budget	\$3,272,723.33
Total FFAR Budget	\$900,000.00
Matching Funder(s)	2Blades, IGI-UC Berkeley
Final Report Date	04/01/2024
Project Director/Principal Investigator Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
- 1.2. How many new jobs were created because of this grant funding?
3
- 1.3. How many jobs were maintained because of this grant funding?
3
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

The ability to make durable and sustainable disease resistance to wheat blast and wheat rusts is extremely important for the food security of this crop as wheat is a major source of

calories in the human diet. This will be especially important as climate change will exacerbate the breakdown of disease resistance which may result in disease pandemics to this important crop. Furthermore, wheat blast has been observed in northern climates as a result of increased temperatures. Furthermore, using genetics to control plant diseases is a more sustainable alternative to chemicals which will reduce the use of fossil fuels

To address this important problem, we are developing genome edited wheat plants for durable and sustainable disease resistance to wheat blast and wheat rusts. Our approach has been to identify orthologous genes in wheat that have been edited for disease resistance in rice that provide durable resistance to a broad spectrum of pathogens. We originally focused our attention on making CRISPR-induced mutations in the wheat TaPi21 and TaBsrk-1 genes based on published results in rice. We did obtain homozygous mutations in both of these genes in all three genomes of wheat, but they did not result in disease resistance. Thus, we have now focused our attention on the wheat dmr6 genes. There are two wheat dmr6 genes and we have obtained mutations in the A, B and D genomes and we are currently selfing these plants. Once we have obtained either bi-allelic or homozygous mutations in all three genomes of both genes we will evaluate these plants for wheat blast and wheat rust resistance. In addition to editing the dmr6 disease susceptibility genes, we are introducing landing pads into the wheat genome by HDR (homology-directed repair). Once the landing pads have been constructed, we will introduce stacks of disease resistance genes using flp recombinase. We have obtained the wheat *Rmg8* gene that confers resistance to many isolates of wheat blast. We have introduced the landing pads into wheat and are currently screening HDR events by PCR.

The ability to make durable disease resistance to these important pathogens of wheat is extremely important for the food security of this crop as wheat is a major source of calories in the human diet. In addition to employing known disease resistance genes we have been investigating novel techniques to engineer NLR genes to detect new pathogen effectors.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Once we have edited plants that disease resistant plant breeders will be able to use these lines in various breeding programs. The agriculture community can utilize our methods for highly variable NLR identification to prioritize breeding lines and genes in cloning. We have provided pre-computed analyses for several crops with Google Colab (see datasets and data management) which can be used by researchers and breeders. Our computational grouping of fungal pathogen effectors can help diagnostic of pathogen races.



- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:
- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
 - b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
 - c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Both our bioinformatic analyses of NLRs and effectors and yeast screens influenced research designs across several other labs, we have openly shared all datasets. Structural clustering of effectors has been utilized across many laboratories working on fungal pathogens. We have also shared our findings and limitations of screens in yeast openly. The postdoc trained on the grant is continuing her work now in the Michelmore lab and applying approaches of NLR-effector interactions she developed in yeast to lettuce genes.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Goal 1. Evaluate CRISPR-Cas9 Gene Edited Wheat Plants for Broad-Spectrum Disease Resistance.

Goal 2. Development of CRISPR-Cas9 Gene Knock-Ins in Wheat.

Goal 3.

Yr1 Screening synthetic NLR library for new effector binding specificities (100% completed).

Yr2 Test synthetic NLRs with new binding specificities for activation of immune response (80% completed)

Yr3. Transform synthetic NLRs into wheat and test for resistance using whole pathogen infection (not completed).



- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Yes, we have pivoted several times in our second goal of optimizing synthetic NLR design to improve recognition and avoid auto-activation. Our project has been greatly advanced through recent breakthroughs in computational modeling of protein structures which we did not envision when submitting this grant and therefore we re-oriented from reliance on solely yeast screens to computationally-informed rational receptor designs.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Major activities:

Our selection of TaPi21 and TaBsrk-1 were based on published results in rice. We did obtain homozygous mutations in both of these genes, but they did not result in disease resistance. Thus, we have now focused our attention on the wheat dmr6 genes. Based on published results in rice we feel that homozygous mutations should produce broad spectrum disease resistance. We have identified and confirmed that wheat contains two dmr6 genes. We have designed guides and harvetested them in vitro to determine if they cut. We have also designed and constructed vectors that contain CRISPR-Cas9 and three different guides. One guide will cut both genes and the other two guides will be specific for a particular dmr 6 gene. In this way, we can ascertain if only one the genes are necessary for function or whether we need to knock out both genes. We have initiated transformation experiments and will get the results in about 6 months.

We have learned that it is important to find a site in the genome that has a high gene editing rate. We have now published that increasing the temperature to 30 degrees C results in high editing frequencies for the Snn5 locus in wheat. The locus encodes a toxin receptor that if present in the plant results in toxin sensitivity. We have obtained high editing rates in this gene when we deliver RNP particles that contain a guide for this gene. Furthermore, introducing a gene into this locus that plant will be become toxin insensitive. In addition, we have obtained T-DNA inserts that contain flip recombinase landing pad. These plants are now growing in the growth chamber and immature embryos will obtained and used for cargo gene

insertions that use flip recombinase. Finally, if we obtain disease resistance dmr6 edited plants, we will attempt to insert the landing pad into the dmr6 gene. In this manner, we could achieve truly durable resistance by not only by introducing disease resistance genes but simultaneously knocking out a disease susceptible gene.

We have been working on both bioinformatics and molecular biology experiments to advance engineering of hypervariable plant NLRs. Our major activity had been to test yeast assay for NLR evolution, in which we tested multiple receptors and their cognate pairs, and confirmed amenability of several NLRs, including Sr35 from wheat to directed evolution in yeast. Our specific objective from this grant which we successfully achieved had been to establish an alternative more sensitive assay than yeast two hybrid assay. We now have split DHFR assay working in our hands with at least 3 NLR platforms. We also produced negative results showing that while NLRs interact well with effectors in split DHFR, the system is not mature enough to allow for directed evolution of NLRs in yeast, mostly due to construct and protein stability in yeast. Bioinformatically, we have successfully predicted 12 key amino acid residues that control receptor specificity in wheat NLR Sr50 and were able to demonstrate specificity transfer to a distant homolog Sr33 by engineering a fully functional synthetic Sr33syn that has 12 amino acids grafted onto its predicted binding site, allowing it to recognize and elicit response to AvrSr50. Furthermore, we identified 2 amino acids at the predicted binding site that are critical for receptor activation (keeping it in ON or OFF state). All current work on Sr33syn has been validated in at least two plant assays: directly in wheat protoplasts as well as in *Nicotiana benthamiana*. We have tested effects of different promoter strengths on activation and auto-activation levels of Sr33syn, and concluded that it would be best to test at least two different promoters in the transgenic lines. Our project has also been revolutionized by advances in computational structural genomics, we have led comparative structural analyses of effectors to prioritize candidates for our designs and provide new information and datasets to research community (Seong and Krasileva 2021, Seong and Krasileva 2023).

Significant results:

- *We have published our proof of principle work on synthetic Sr33 design (Tamborski et al 2023). We have advanced this project further by successfully designing synthetic Sr50 variant that can recognize and respond to virulent AvrSr50 effector that escaped recognition (unpublished).*
- *We have encountered significant technical challenges working with Orthorep mutator system in yeast when using full length NLRs. This was likely due to instability of large constructs containing the full-length gene (3 kb). We concluded that shorter LRR only constructs or non-repetitive effector-binding NLR domains would be more suitable for future directed evolution of NLRs in yeast.*
- *We have finished and published comparative structural genomics work (Seong and Krasileva 2023) where we identified structurally similar effectors, including those similar to AvrSr35 and AvrSr50 that are prime targets for synthetic Sr50 and Sr35. We have synthesized these structurally similar effectors and utilized split DHFR yeast system that we established for Sr50/AvrSr50 and Sr35/AvrSr35 to design and screen synthetic constructs against structurally similar effectors.*



- *We have established in house highly efficient wheat transformation in the Krasileva lab and began transforming our synthetic Sr33 and Sr50 constructs in the susceptible wheat varieties for whole pathogen testing. Whole pathogen testing and stacking NLRs will be completed beyond the time of this grant.*

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

We maintain detailed electronic notes on Benchling. All students and postdocs in the lab work with hypotheses tables to identify critical experiments and necessary controls for both bioinformatic and lab work.

- 2.8. List key stakeholders that could be served by results of this project.

Academic community
Plant breeders

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Graduate students, postdocs and PIs have presented this work at academic conferences and invited seminars. Work in the Krasileva lab has been highlighted by The Innovative Genomics Institute through an interview (<https://innovativegenomics.org/news/meet-an-igi-scientist-ksenia-krasileva/>). The College of Letters and Science highlighted our work in their most recent Breakthroughs article distributed widely to all College alumni. Krasileva lab had put together "Genomes and Plant Health" outreach workshop for high school students and undergraduates. Krasileva has been advising development of new curricula in Genomics and Computational Biology for High School students with Biotech Partners

(<http://www.biotechpartners.org>).

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Our main challenge has been instability of full length NLRs when they are integrated into Orthorep mutator system cassette with yeast losing inserted genes. Since NLRs are large genes and the system has been previously tested for only smaller constructs (~1kb), we have pivoted several times. Since construct integration in this system requires homologous recombination, our main conclusion has been to utilize non-repetitive NLR domains in future work.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Each year, Krasileva lab organizes a dedicated three day forward planning symposium, which provides yearly opportunities to form and discuss Individual Development Plans. We provide structured activities to all trainees across several areas of professional development. Throughout the year, we have regular meetings and additional opportunities for professional development. All trainees attended at least one national and one international conference a year and present their research as talks or posters. Graduate student involved in this project received formal interdisciplinary training facilitated by UC Berkeley Designated Emphasis in Genomics and Computational Biology program.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications,



reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

scholarly publications
video
software
datasets
curricula

3.2. Please provide a list of citations for the information products produced during the project.

- **Code** for identification of hvNLRs is available on GitHub ([link](#)).
- **Code** used for wheat NLR ligand binding site prediction in Tamborski et al study: https://github.com/krasileva-group/Sr33-Sr50_analysis
- **CoLab** enabling community to analyse their own receptor and predict ligand binding site using our method: <https://colab.research.google.com/github/daniilprigozhin/NLRCladeFinder/blob/main/NLRCladeFinder.ipynb>
- **Dataset** for "Intraspecies diversity reveals a subset of highly variable plant immune receptors and predicts their binding sites" Zenodo DOI [10.5281/zenodo.3951780](https://doi.org/10.5281/zenodo.3951780)
- **Dataset** for "Comparative computational structural genomics highlights divergent evolution of fungal effectors". Zenodo DOI: <https://doi.org/10.5281/zenodo.6480453>
- **Curricula** for Genomes and Plant Health: <https://krasilevalab.org/outreach/>
- **Video** recording of talks: <https://krasilevalab.org>

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Poddar, S.J. Tanaka, K. Running, K. Kariyawasam, K. Gayan, J Faris, T. Friesen, M. Cho, J Kate, B. Staskawicz. Frontiers in Plant Science 2023. Optimization of highly efficient exogenous-DNA-free Cas9-ribonucleoprotein mediated gene editing in disease susceptibility loci in wheat (*Triticum aestivum* L.) Frontiers in Plant Science 2023. doi: [10.3389/fpls.2022.1084700](https://doi.org/10.3389/fpls.2022.1084700)

Tamborski J, Seong K, Liu F, Staskawicz B, Krasileva K.V "Engineering of Sr33 and Sr50 plant immune receptors to alter recognition specificity and autoactivity" Mol Plant Microbe Interact. 2023 Mar 3. doi: 10.1094/MPMI-07-22-0154-R.

Seong K, Krasileva KV. "Prediction of effector protein structures from fungal phytopathogens enables evolutionary analyses". Nat Microbiol. 2023 Jan;8(1):174-187. doi: <https://doi.org/10.1038/s41564-022-01287-6>.

Seong K, Krasileva KV "Computational structural genomics unravels common folds and predicted functions in the secretome of fungal phytopathogen *Magnaporthe oryzae*" Mol Plant Microbe Interact. 2021 Aug 19. (bioRxiv Jan 2021) <https://doi.org/10.1094/MPMI-03-21-0071-R>.

- The top MPMI Graduate Student Paper Award for 2021

Prigozhin DM, Rangavajjhala S, Krasileva KV "Majority of the highly variable NLRs in maize share genomic location and contain additional target-binding domains" bioRxiv 2022.10.05.510735; doi: <https://doi.org/10.1101/2022.10.05.510735>

Prigozhin DM, Krasileva KV. Analysis of intraspecies diversity reveals a subset of highly variable plant immune receptors and predicts their binding sites. Plant Cell. 2021 May 31;33(4):998-1015. doi: 10.1093/plcell/koab013.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

<https://krasilevalab.org>

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	WO/2021/113569

Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)
- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.
- **Code** for identification of hvNLRs is available on GitHub ([link](#)).
 - **Code** used for wheat NLR ligand binding site prediction in Tamborski et al study: https://github.com/krasileva-group/Sr33-Sr50_analysis
 - **CoLab** enabling community to analyse their own receptor and predict ligand binding site using our method: <https://colab.research.google.com/github/daniilprigozhin/NLRCladeFinder/blob/main/NLRCladeFinder.ipynb>
 - **Dataset** for "Intraspecies diversity reveals a subset of highly variable plant immune receptors and predicts their binding sites" Zenodo DOI [10.5281/zenodo.3951780](https://doi.org/10.5281/zenodo.3951780)
 - **Dataset** for "Comparative computational structural genomics highlights divergent evolution of fungal effectors". Zenodo DOI: <https://doi.org/10.5281/zenodo.6480453>



- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Yes, we deposited code on github and related datasets produced by our analyses on Zenodo.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Krasileva laboratory maintains electronic notebooks on Benchling.